Supplementary Video 1 | Confidence map generated by CELLECT on a dataset of membrane-labeled cells. We captured the dynamic migration of neutrophils in the spleen using intravital imaging with scanning light-field microscope[4] (left). From this dataset, we generated and visualized confidence maps with and without CEN using CELLECT (middle and right). The confidence maps generated by CELLECT provide a clearer visualization of cell regions and their movement dynamics. The standard confidence map (without CEN) captures a broader area of the cell regions (middle), while the center-enhanced confidence map (with CEN) highlights only a small, concentrated portion of the cell centers (right). CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset.

Supplementary Video 2 | Efficient cell tracking of CELLECT on *C. elegans* dataset compared with Linajea. Left: MIP of the publicly available dataset (Mskcc-confocal) showing the *C. elegans* embryo development process. Middle: Visualization of cell tracking results using CELLECT. Right: Visualization of cell tracking results using Linajea. The initial 4 seed cells are marked with distinct 4 colors. Differentiated cells are assigned the same color as their mother cell, illustrating lineage tracing capability. CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset.

Supplementary Video 3 | 3D cell tracking of large-scale B cells during the formation of germinal center. Upper left: Orthogonal MIPs of B-cells dynamics in mouse lymph node during germinal center formation captured by 2pSAM. Upper middle: Short-term traces of B-cell migration over the last 10 minutes. Upper right: Continuous traces of B-cell migration. Bottom: Visualization of continuous tracking of a single cell dividing into 2 cells, as obtained by CELLECT and Imaris, respectively. CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset.

Supplementary Video 4 | Continuous tracking of neural activities at single-cell resolution in *Drosophila* brain during tissue deformation. a, MIP of neural activities in *Drosophila* brain with tracking results obtained by CELLECT, Imaris, and TrackMate-Weka for comparisons. **b**, Evaluation of long-term tracking performance by tracking cells from the first frame. Fully tracked cells are shown in green, while partially tracked cells are shown in red. **c**, Heatmap of the neural activities of tracked cells over time. **d**, Representative results of a single neuron tracked by 3-different methods (CELLECT, Imaris and TrackMate-Weka) for comparisons. **e**, Demonstration of multiple cells tracked by CELLECT. 5 cells are depicted with 5 different colors. Newly detected cells after the first frame are not visualized or analyzed in this study. The calcium traces shown in (**c**-**e**) are normalized. The areas corresponding to (**d**) and (**e**) are depicted with blue and purple squares in the MIP shown in (**a**). CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset.